

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Spring-2025, Midterm Exam

Course Code: CSE-3529, Title: System Analysis and Design

Full Marks: 30, Time: 1 Hour 30 Minutes

Answer all of the following questions

- | | | Marks | |
|----|---|-------|-----|
| 1. | a) Define information. Discuss the key qualities that make information valuable and effective in decision-making. | 5 | CO1 |
| | b) A software development team is working on a complex project with high-risk factors and frequent changes in requirements. The project manager decides to divide the development into smaller, manageable phases, each with risk assessment, prototyping, and client feedback. After each phase, the team reviews the progress, adjusts plans, and moves on to the next iteration.

What software development model is being used in this scenario, and how does it address the challenges of risk management and iterative development? | 5 | CO3 |
| 2. | a) What is a system? Discuss different types of information systems and their roles in organizations. | 5 | CO1 |
| | b) What is the Systems Development Life Cycle (SDLC), and how does it help in building an information system? Explain the key phases of the SDLC and provide an example of how each phase might be applied in the development of an e-commerce platform. | 5 | CO1 |
| | OR (of 2b) | | |
| | ✓ Which SDLC model would you adopt for developing an e-commerce platform that requires small and there are no uncertain requirements? Justify your answer reasoning with the advantages and disadvantages of your model. | 5 | |
| 3. | a) Discuss both internal and external factors that affect system project. | 5 | CO1 |
| | b) Explain the role of SWOT analysis in the strategic planning process for IT projects. How does a SWOT analysis help IT managers in identifying strengths, weaknesses, opportunities, and threats? Provide examples of how each element can impact IT project decisions | 5 | CO3 |
| | OR (of 3b) | | |
| | Describe the steps to conduct the economic feasibility analysis with proper example. | 5 | |

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Examination, Spring 2024

Course Code: CSE-3529

Course Title: System Analysis and Design

Total marks: 30, Time: 1 hour 30 minutes.

[Answer all the questions. Figures in the right-hand margin indicate full marks]

1. a) Define system. Discuss the stages in building system. Who is system analyst? Explain role, responsibilities and qualities of a system analyst. 5 CO1
- b) TechGadget Zone, a prominent electronics retailer, has announced a special promotion named "TechFiesta Mega Sale," offering a 30% discount on all electronic gadgets for a limited time, starting from 1st January to 15th January 2024. "How does the TechFiesta Mega Sale promotion at TechGadget Zone exemplify the integration of strategic, tactical, and operational information? Provide specific details on how the strategic decision-making, the tactical planning and the operational considerations addressed to ensure a smooth and successful implementation during the specified period." 5 CO2
2. a) Discuss Rapid Application Development (RAD) and Joint Application Development (JAD) methodologies with their advantages and disadvantages. 5 CO1
OR (of 2a)
What is SDLC? Briefly describe the different phases of SDLC.
- b) A software development company currently offered to develop following projects: 5 CO2
 - i. A new accounting software for a client. The client has provided detailed requirements upfront and expects the software to be delivered in a short period of time.
 - ii. Another project is a new social networking platform. The company wants to adapt to changing market needs quickly and release new features iteratively to gather user feedback.Above the scenario which SDLC model would adopt for developing both projects? Justify your answer.
3. a) Define project management. Describe the execution phase of project management. 5 CO1
- b) A very basic cash flow projection for an IT Projects is, in the Year 0,1,2,3 costs are 50000, 8000, 10000, 14000 and benefits are 0, 25000, 30000, 48000. Find out the followings for the above conditions: 5 CO2
 - (i) Return on Investment
 - (ii) Break-Even Point

OR (of 3b)

Describe cost-benefit analysis. Briefly describe the different metrics to measure projects feasibility in terms of economic value.

International Islamic University Chittagong
Department of Computer Science and Engineering

B. Sc. in CSE Mid-term Examination, Autumn 2023

Course Code: CSE 3529 Title: System Analysis and Design

Total marks: 30 Time: 1 hour 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

Course Outcomes (COs) of the Questions	
CO1	Understand project management and system development life cycle.
CO2	Analyze the requirement and feasibility of software application
CO3	Design an information system.
CO4	Compile the system design document

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analysis	Evaluate	Create

- 1.a) "Foodpanda" is a popular food delivery app that allows users to order food from various restaurants. Considering the following statements aligned with the app, classify them into *Strategic*, *Tactical*, and *Operational* type of information. Justify your answer. 5 CO2 DL U
- i. In-app pop-up notifications (immediately when the user opens the app) highlighting the new features
 - ii. Added in-app features of "top 10 most ordered meals of the week", and "top rated restaurants near you"
 - iii. Showing trendiest feature in app, i.e. calorie count for every item
- b) Write characteristics of quality information. 5 CO3 C
- Suppose an online learning platform seeks to enhance its user engagement and course completion rates. So, a system analyst designs lucrative features such as progress tracking, interactive quizzes, and discussion forums to promote student involvement. What do you think what kind of roles/responsibilities can a system analyst play in such cases according to the above scenario?

- 2.a) Which SDLC model would you adopt for developing an e-commerce platform that requires small and there are no uncertain requirements? Justify your answer reasoning with the advantages and disadvantages of your model. 5 C01 U

Or,

Suppose you want to develop an online learning platform. It develops modules on various subjects, enabling instructors to create and deliver content, and students to engage in interactive learning. Regular feedback from educators helps refine the platform's features and usability. Which SDLC model will be more suitable for the above learning platform? Justify your answer

- b) Distinguish between Methodologies, Models, Tools and Techniques with example. 5 C01 U

- 3.a) What is the significance of feasibility analysis in project management? Which phase of project management requires feasibility analysis? Briefly discuss about three major areas to conduct a feasibility study of an organization. 5 C01 U

- b) Describe the steps to conduct the economic feasibility analysis with proper example. 5 C02 An

Or

Define the following:

- i) Deliverables
- ii) Break-Even Point (with mathematical explanation)
- iii) Net Present Value (with mathematical explanation)
- iv) Recurring cost
- v) Return on Investment (with mathematical explanation)

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring 2023

Course Code: CSE 3529

Course Title: System Analysis and Design

Total marks: 30

Time: 1 hour 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

Course Outcomes (COs) of the Questions	
CO1	Understand project management and system development life cycle.
CO2	Analyze the requirement and feasibility of software application
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Bloom's Levels of the Questions						
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CO DL

1. a) Suppose, you are a project manager of the Nascenia software firm in Bangladesh that currently intend to hire a system analyst. Now, write a draft comprising the skills and experience that an ideal system analyst should have. 3 CO2 U
- b) Rokomari.com is one of the popular online book-selling sites in Bangladesh. They recently introduced an offer stating that "Rokomari Online Book Fair offer a 25% discount on every book for 1 month, starting from 1st February to 28th February 2023." From the viewpoint of Strategic, Tactical, or Operational information types, analyze which kind of information this offer belongs to. Justify your answer. 3 CO3 An
- c) What are the types of information systems? Consider that Y wants to reach his/her university campus quickly from Chattogram City. He/She uses GPS route planning system that plans the fastest and best routes between two points analyzing available options. What kind of information system does it fall into? Justify your answer. 4 CO3 An

2.

- a) Which SDLC model would you adopt for developing an e-commerce platform that requires constant adaptation to user feedback and market trends? Justify your answer with a comparative study with other models. 5 C01 An

Or

Suppose you want to manufacture and deliver a software program that determines which university will be best suited for a student opting for an Undergraduate Degree based on student scores and university rankings.

State the SDLC model that you will choose for this above mentioned software program. Justify your answer reasoning with the advantages of your model over other existing models.

- b) Discuss RAD and JAD methodologies with their advantages and disadvantages. 5 C01 U

Or

What is SDLC? Briefly describe the different phases of SDLC.

3.

- a) What is the significance of feasibility analysis in project management? Which phase of project management requires feasibility analysis? Briefly discuss about 3 major areas to conduct a feasibility study of an organization. 5 C01 U

- b) What is Break-Even Point analysis? Explain with mathematical equations and a necessary figure. 5 C02 U

Or

Define the following: i) Tangible cost ii) Intangible cost iii) SWOT Analysis iv) Recurring cost v) Return on Investment

International Islamic University Chittagong (IIUC)

Dept. of Computer Science & Engineering (CSE)

Midterm Examination: Spring 2022

Program: B.Sc. in CSE

Course Code: **CSE-3529**

Time: 1.5 Hours

Course Title: **System Analysis & Design.**

Full Marks: 30

Answer all the *three* questions. All parts of a question must be answered sequentially. Figures in the right margin indicate full marks.

COL2 1.(a) What are various fact finding techniques? Discuss the advantages and Disadvantages of using of these techniques. 4

COL2 (b) Define System Prototype. Explain System Prototype Method (SPM). What are the features of prototyping. 4

OR

COL1 (c) Discuss Linear Sequential VS Waterfall model with feedback
Describe system and information system with example. Why computer based information system is needed? 2

COL2 2.(a) What is the System Development Life Cycle? Discuss the steps of SDLC in details. 4

COL1 (b) Define the following terms: 4
(i)Boundary & Environment
(ii)Sub system
(iii)Black box
(iv)Feedback Control

OR

COL1 (c) What is System? What are the various components of System? Describe the different types of System.
Consider that you need to build online library management software for IIUC.To develop this software which techniques do you prefer from direct observation or document analysis. Justify your answer. 2

COL3 3.(a) Write short notes on: 4
(i)Breakeven Point (ii)Return on Investment
(iii) Net Present Value (iv)Profitability Index

COL3 (b) What is feasibility study? What are different types of feasibility study? Write different steps of Feasibility analysis. 4

COL3 (c) Differentiate between System analysis and design. 2

OR

Difference between RAD & JAD.